# Technology Innovation HUB Program Update

For the **National Football League**
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Program Update

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# Meeting Agenda

1. **Background**
   Recap of Previous Presentation
   Explanation of Value Zone

2. **Innovation HUB**
   Program Design and Criteria
   Governance Structure

3. **Operating Model of the Innovation Hub**
   Execution Phase Gates

4. **Pilot Project Updates**
   Optical Tracking Progress
   NFL Player View Project Developments
   NLP / Multi-modal AI Milestones

5. **NLP / Multi-modal AI Pitch**
   Introduction
   Business Value

# 4 Zones Model: Business Value Creation

| | DISRUPTION INNOVATIONS | SUSTAINING INNOVATIONS |
|---|---|---|
| **NFL Goals & Focus Area** | **Transformation Zone** | **Performance Zone** |
| **Enabling Investments** | **Incubation Zone** | **Productivity Zone** |

- → **Disruptive Innovations:** New business models and operating models
- → **Sustaining Innovations:** Extension and improvements to already existing innovations inside the organizations

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# Program Design & Criteria

## Program Design

- ➔ VC Model of Investment
- ➔ Full visibility on objectives and progress of each project
- ➔ Centralized decision making on which projects to fund

## Established Success Criteria

- ➔ Initial funding decisions based on rigorous screening and evaluation process
- ➔ Control of the funding is linked to stage gates or milestones
- ➔ Flexible exit options based on iterative learning and market conditions

# Governance Structure

The Governance Committee's mission is to steer the Innovation Hub program through collaborative design, guiding league-wide decisions for maximum impact.

**Executive Steering Committee**
EVPs

**Program Execution Committee**
SVPs - Information Technology, Player Health and Safety, Football Operations, NFL Media, Data & Analytics, Finance, and Sponsorship

**Innovation Tech Advisory Committee**
NFL Technical Working Group

**Innovation HUB PMO**
Intake, Prioritization, & Program Management

# HUB Execution Gates or Operation Model

### Intake

- Internal/External Idea Submission
- **NFL Business Value**

### Concept

- **Zone Placement**
- Depth Evaluation
- Technical Feasibility

### Business Case

- Product/Service definition
- **Business Justification**
- Development Plan

### MVP

- Develop prototype to verify and validate
- **Test market fit**
- Several iterations

### ROI

- Financial viability
- **Implementation strategy**
- Total Cost of Ownership

### Grow

- **Transition to funded operational project**

Stage 1

Stage 2

Stage 3

Stage 4

Stage 5

**HUB Funded Projects**

# POC Projects

1. Optical Tracking
2. NFL Player View
3. Multi-Modal AI

# Project Stage Placements

| Intake | Concept | Business Case | MVP | ROI | Grow |
|---|---|---|---|---|---|
| **Multi-modal AI**<br>Brad Boim | **NFL VIEW**<br>Marc Hodakowski | | | **Optical Tracking**<br>Aaron Amendola | |

Stage 1 — Stage 2 — Stage 3 — Stage 4 — Stage 5

**HUB Funded Projects**

# Optical Tracking

ROI

**Overall Scope:** The optical tracking scope aims to identify an innovative solution for line-to-gain detection and other real-time officiating use cases.

**Current State:**

- **Conducting 3 POCs to determine the accuracy of each demo. Vendors:** HawkEye, ChyronHego, Genius/Second Spectrum
- **Stadium Evaluations**: Each vendor is undergoing evaluations and testing in stadiums upon approval
- **Vendor Scorecard & Partner Screening**: A vendor scorecard has been developed to assess compatibility and are actively screening potential partners during the Proof of Concept (POC) phase

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# Optical Tracking, Cont.

## Use Case

- **Line-to-Gain**: Utilize optical tracking technology to improve the accuracy of line-to-gain detection
- **Game Presentation:** Optimized data utilization for all games (Including in-stadium and broadcast executions)

## Next Steps

- Data accuracy comparison among vendors, aiming for consistently achieving ½ inch accuracy or better ( ½ or less)
- Identify the vendor that best aligns with the goals identified through RFI Vendor Evaluation Scorecard

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# Optical Tracking

## Vendor Assessment

| | **HawkEye**<br>*Current Vendor* | **Genius/Second Spectrum**<br>*Current Vendor* | **ChyronHego**<br>*New Vendor* |
|---|---|---|---|
| **Overview** | Technology is currently deployed in 3 stadiums (LV, MIA, NY) | SoFi Stadium (LA) | Technology has been deployed at 5 stadiums for MLS soccer (ATL, CAR, CHI, NE, SEA) |
| **Update** | Data from testing has been analyzed | Installation has been completed in SoFi Stadium (LA) | NDA has been signed |
| **Next Steps** | A review with the Competition Committee  and the Future of Football Committee scheduled during Combine (Feb 27 - Mar 1) | Tech evaluation scheduled for the January 7th game | Validating solution internally before enabling system for January 7th |

# NFL View

Concept

**Overall Scope:** NFL View's primary emphasis is player health and safety. Their concentration encompasses player-related elements such as player speed, impact force and injury prevention.

**Current State:**

- Enhance data collection for player health and safety.
- Three years of testing and utilizing original Intel cameras for player movement tracking.

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# NFL View

## Use Case

- **Player Health and Safety**: Proactively preventing injuries by identifying game events that enhance player safety.
- **Identifying Game events:** Identifying game events can help improve player engagement and enhance the overall gaming experience.
- **Skeletal tracking**: This is implemented to proactively enhance player safety and performance.

## Next Steps

- Collaborating with the PHS team to create a comprehensive plan for business testing, cost analysis, and project timeline development.

# Multimodal AI Pitch

Brad Boim

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# Multi-Modal AI

Intake

**Overall Scope:** The project scope focuses on developing a high-level AI-powered media search tool for media assets within our core Media Asset Management (MAM) system. This will be done during the 2023 Season Playoffs

## Strategic Alignment

- New Technology
- Non-Game Media
- International Expansion

## POC already completed – June thru October

100 hours of NFL content indexed, 10 taxonomies fine-tuned

## Phase One Goals – April 2024

- Direct integration of TwelveLabs API with Vidispine MAM
- Build, test and refine integration and validate results
- Identify and engage power users to test and provide feedback

# Innovation Hub - Funding Request

*Designed for:* MutliModal AI

*Designed by:* Meloni Boatswain

*Date:* 12/12/2023

*Version:* 1.1

## Problem

- Current media search process is inefficient and time consuming
- Most content is not logged, buried deep in the archive, and impossible to find.
- Human logging process is antiquated, costly and error prone ($25K / year)

## Solution

***Develop an automated solution leveraging Multi-Modal AI via the TwelveLabs API to allow Natural Language search across our content library at a scale.***

## Unique Value Prop

- Develop a natural language (AI/ML) search tool for media assets within the NFL to unlocking previously unavailable content
- Train model for a customized taxonomy for exclusive use by the NFL

## Expected Outcome

- Faster search efficiency
- Improved discovery of our media archive, uncovering assets previously buried in the archive
- Quick adoption, as it utilizes UI tools already in use

## Customer Segments

- Internal Content Creators
- NFL Media Producers
- Complience

## Existing Alternatives

- Manual asset tagging
- Not logging content at all
- Traditional AI methods

## Key Metrics

- Accuracy of search results
- Search time reduced from minutes/hours to seconds
- Is the indexing pipeline sustainable?
- Faster user adoption

## High-Level Concept

- TwelveLabs multi-modal AI extracts multiple features from video such as time, objects, text in video, conversation, faces, and actions, synthesizing this information into vectors.
- Vectors enable fast, semantic and scalable search using natural language.

## Next Steps

- Direct integration into the NFL's existing MAM system
- Develop indexing and search pipeline
- Fine-Tune NFL Taxonomies
- Deployment and testing during 2023 Playoffs to determine capabilities and benefits

## Early Adopters

- NFL Media Teams (L.A., NJ, NY)
- NFL Game Operations

## Cost Structure

Integration, development, testing, and initial deployment:

- ARVATO SYSTEMS NORTH AMERICA $15,000
- TwelveLabs $36,300

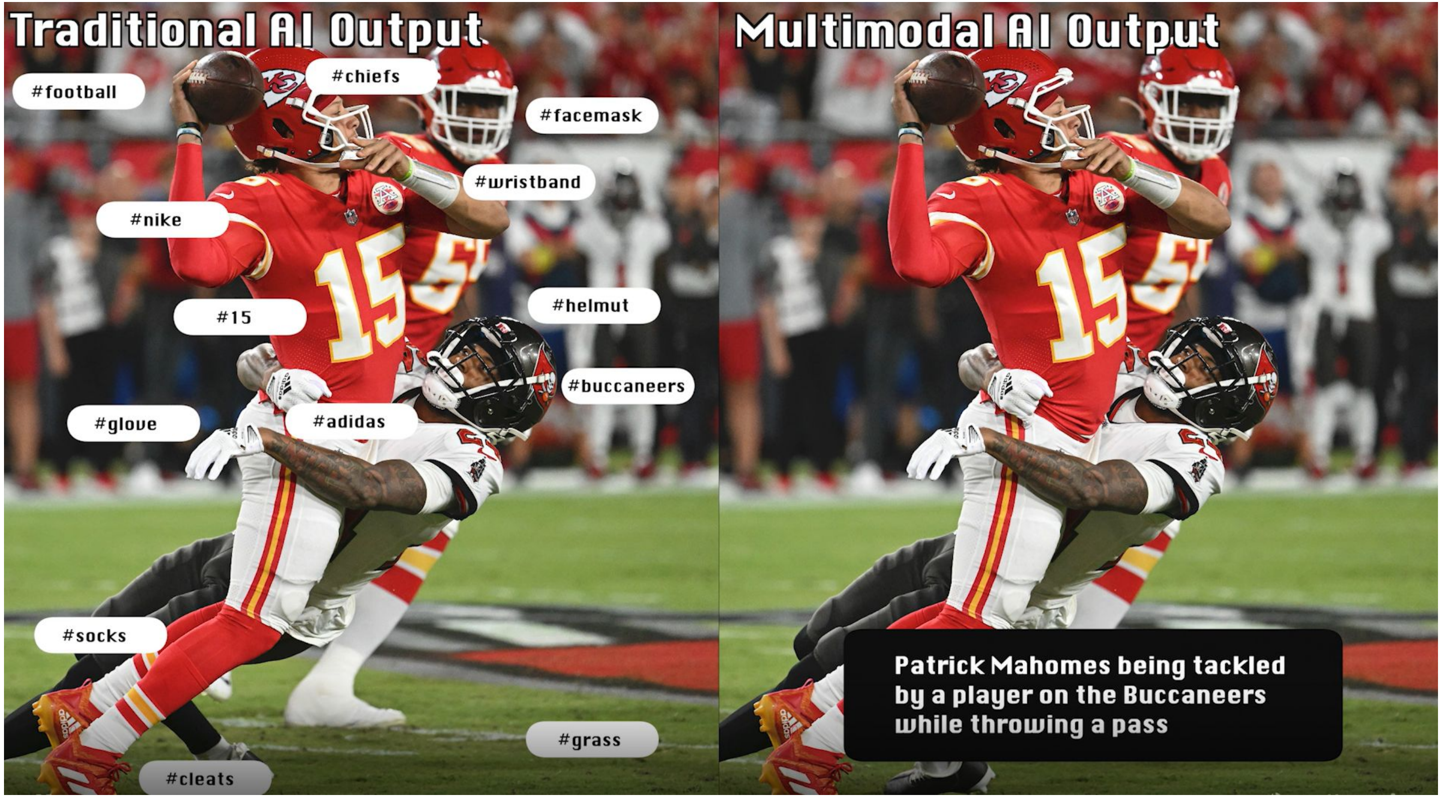
**Total: $51,300**

## Value Proposition

- Cost savings from increased efficiency and accuracy.
- Exploration of Gen AI technology targeting potential applications within the NFL.
- Gaining leverage from existing investments in our media asset library utilizing emerging Gen AI technology.

# Multi-Modal AI

## Video Demo



# Pitch Feedback

# Next Steps

# STRATEGIC PLANNING FOCUS

League Goals: Increase Consumption | Fan Growth | Grow Key Demographics

## Focus Areas

**Direct-to-Consumer**

NFL SUNDAY TICKET

NFL+

**International**

**Legalized Sports Betting**

OFFICIAL NFL DATA

**Non-Game Media Content**

NFL FILMS

**New Technology**

Blockchain

NFT

AR

## Foundational Investments

**Game**

Quality

Youth

NFL FLAG

PH&S

BUILDING A BETTER GAME

Digital Athlete

**Fan Development**

Avid

Youth / Latinx

**Business Operations**

DEI

INSPIRE CHANGE NFL

Data/Tech

Fan 1:1

*League goals in no particular order